

Atama
Team 20 Product Backlog
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Problem Statement

There are a lot of applications to support studying, such as Quizlet, Anki, and RemNote, but they all have differing and scattered features, lack standardization and organization, and are often blocked or limited behind ads and paywalls. We want to make a centralized study application with quality tools focused on deep learning, make study materials available in one spot, and encourage student interaction and collaboration. With tools like flashcards, study guides, and practice tests supplemented with generative AI, as well as a shared resource drive and study group matching, students can now do all of their studying in one streamlined app.

Background Information

Audience

As online platforms and services become the norm for study materials and tools, students often need to rely on multiple different services to access their desired breadth of materials and tools. Existing options often rely on premium or paywalled features and lack the centralization that students need.

Similar Platforms

There are several existing study tools, such as Quizlet, Anki, Remnote, and Studocu, that allow students to use study tools in different ways. Quizlet, Anki, and Remnote focus on flashcards with different variations, while Studocu allows students to pool study materials.

Limitations of Similar Platforms

The main limitation of these platforms is that useful features are behind paywalls (Quizlet, Remnote, Studocu), making it difficult for users with limited finances to use these platforms. In addition, the main study tool is only flashcards (Anki, Remnote), the application runs off a give-one-get-one or subscription model (Studocu), and there lacks consolidation or organization of combined study materials in a school or class (Quizlet). Atama will include study tools like games and practice tests, as well as provide a shared drive of study materials that students contribute to. This ensures a wide variety of available study tools, all in one location and entirely free for all users. The requirement to access the shared drive for a university course will simply require a school email with the account, instead of requiring a user to pay or upload files in order to see material. Additionally, the pages of courses at specific universities will allow students to easily navigate to relevant study materials for their classes.

Functional Requirements

1. As a user, I would like to register for an account.
2. As a user, I would like to log in and manage my account.
3. As a user, I would like to change my username, password, privacy settings, and profile picture.
4. As a user, I would like to toggle dark/light mode.
5. As a user, I would like to view my library of study materials I created and allow others to view my library if I set it to public.
6. As a user, I would like to create and edit folders in my library of study materials for organization purposes.
7. As a user, I would like to opt out of or into AI-generated study content.
8. As a user, I would like to create flashcard sets.
9. As a user, I would like to study flashcard sets.
10. As a user, I would like to be able to create different types of flashcards, such as drag and drop labels onto a diagram, putting steps in order, and fill-in-the-blank.
11. As a user, I would like to access and use other users' flashcard sets to study.
12. As a user, I would like to rate others' flashcard sets with a star system.
13. As a user, I would like to review others' flashcard sets with a tagging system similar to that of RateMyProfessors with tags such as "Quick to study", "All class content", "Chapter-specific", "Professor-specific", etc.
14. As a user, I would like to externally share flashcard sets and other study materials through a generated shareable link.
15. As a user, I would like to upload my notes or slideshows in PDF form so that the application generates flashcards for me, using AI if enabled. Otherwise, formatted text can also be used to create flashcards.
16. As a user, I would like to have the application generate practice tests from my flashcard set(s) or other study materials, using AI if enabled. Otherwise, formatted text can also be used to create practice tests.
17. As a user, I would like to download or export my flashcard sets and practice tests.
18. As a user, I would like to create a concept map using my flashcards through drag-and-drop.
19. As a user, I would like the application to generate a concept map for me using generative AI.
20. As a user, I would like to choose between different studying tools, like flashcards and practice tests.
21. As a user, I would like to study using individual games such as memory.
22. As a user, I would like to study using collaborative games structured like Kahoot.
23. As a user, I would like flashcard sets and study materials to be suggested to me based on the topics I am studying.

24. As a user, I would like to filter searches by school, class, type of study material, rating, creation date of study material, etc.
25. As a user, I would like to search for and join my school.
26. As a user, I would like to manually join specific classes in my school for the semester.
27. As a user, I would like to upload my schedule and automatically join specific classes in my school for the semester.
28. As a user, I would like to automatically leave classes once the semester is over.
29. As a user, I would like student verification for joining a school.
30. As a user, I would like to search for content for specific classes at my school.
31. As a user, I would like to upload, access, and download study materials in the shared drive for my specific class.
32. As a user, I would like to filter searches in the shared drives to find content more easily.
33. As a user, I would like to join study groups for my specific classes.
34. As a user, I would like to participate in and observe leaderboards regarding how much time members of my study group spend practicing, or other statistics.
35. As a user, I would like to send nudges (via notifications/emails) to members in my study group to encourage them to study or practice.
36. As a user, I would like to set study schedules and goals so that I know when and how much to practice.
37. As a user, I would like to receive desktop notifications that tell me when to start studying based on the study schedule from the preceding user story.
38. As a user, I would like to observe and maintain a study streak.
39. As a user, I would like to set an exam date and have a countdown as a reminder.
40. As a user, I would like to view the progress I have made in studying a set and how much time I have spent on it.
41. As a user, I would like a built in timer to utilize study time management methods, such as Pomodoro.
42. As a user, I would like to report content that is inappropriate for the purpose of the application.
43. As a user, I would like to be able to migrate study materials from other platforms (e.g., Quizlet flashcards) to this platform.
44. As a user, I would like to receive feedback from an AI-study buddy on how to improve my answers. (if time allows)
45. As a user, I would like to chat with my study buddies in-app to streamline communication. (if time allows)
46. As a user, I would like to video conference my study buddies in-app to streamline communication. (if time allows)

Non-Functional Requirements

Architecture and Performance

Our application will have a separate frontend and backend. The backend will be Java (Springboot)-based and make relevant API calls. We will also use Amazon S3 for our database to store and manage our system and user data, including the uploaded study materials, for scalability and developer usability. The frontend will be JS (React)-based. We currently plan to use Gemini for AI features.

Usability

The user interface should be easy to navigate and understand. Ideally, users will be able to intuitively access and use the site upon first landing.

For response time, the user should be able to upload or download materials from the shared drive at a speed relative to their internet speed and comparable to Google Drive. We will aim for a download speed of 50-100 Mbps and an upload speed of 2-10 Mbps. Furthermore, the storage space given to a university course's shared drive should hold up to 15 Gb, while personal user libraries should hold up to 2 Gb. The search bar should have a response time of 800 ms or less.

Security

Users who are logged out or have no account will only be able to view public flashcard sets. These users will also not be able to create flashcard sets and study materials, or contribute to any shared drives. Furthermore, the personal information of all users, such as their email and college affiliation, will not be visible to users who are not logged in.

If a logged-in user wants to access the shared drive of a particular class of a university, they must be a verified member of that university. There will be privacy settings so that members can set their profiles, flashcards, and study materials to public or private view.

Should they have security concerns with generative AI chatbots, users who do not wish to use generative AI features will have the option to toggle them in their settings.

Deployment/Hosting

We currently plan to deploy our application using GitHub Pages for the frontend and Vercel to deploy the Java backend.